CHAPTER 3

Interaction of Light with Matter

3.1 Macroscopic Aspects for Solids

3.1.1 Boundary Conditions

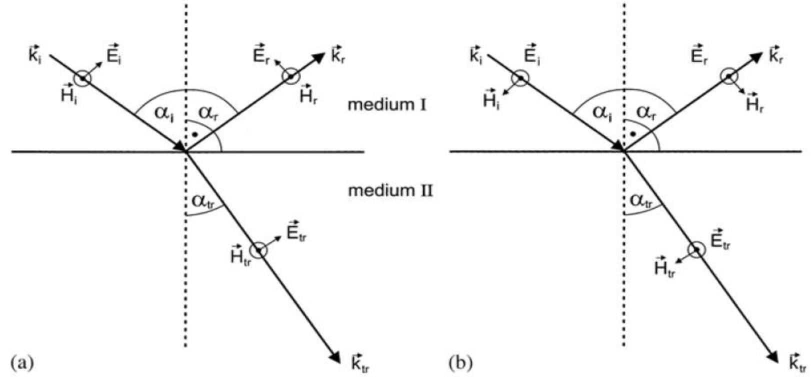


Fig. 3.1. The $\boldsymbol{E}$ and $\boldsymbol{H}$ fields and the wave vectors $\boldsymbol{k}$ for incident, transmitted and reflected beams at an interface between two isotropic media for two different, orthogonal, linear polarizations (**a**) and (**b**) respectively

- Assume :1. media I and II are isotropic ; 2. there is only one reflected and one transmitted beam
- The coefficients of reflection r and transmission t of the interface :
- For Fig.3.1a :

$$r_{\parallel} = \frac{E_{0r}}{E_{0i}} \qquad t_{\parallel} = \frac{E_{0tr}}{E_{0i}}$$

- For Fig.3.1b:

$$r_{\perp} = \frac{E_{0r}}{E_{0i}} \qquad t_{\perp} = \frac{E_{0tr}}{E_{0i}}$$

- For measurement , reflectivity R and transmittivity T of an interface for the intensities:

$$R_{\perp,\parallel} = \left|r_{\perp,\parallel}\right|^2$$

? How to calculate them?

1. Maxwell's equations with boundary conditions
2. The laws of Gauss and Stokes

$$\int_{volume} \nabla\cdot\mathbf{A}\,d\tau = \oint_{surface} \mathbf{A}\cdot d\mathbf{f} \qquad \int_{surface} (\nabla\times\mathbf{A})\cdot d\mathbf{f} = \oint_{line} \mathbf{A}\cdot d\mathbf{s}$$

$$\int_{volume} \nabla\cdot\mathbf{A}\,d\tau = \oint_{surface} \mathbf{A}\cdot d\mathbf{f}$$

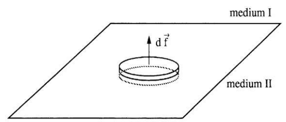


Fig. 3.2 Schematic drawing of the infinitesimally small cylinder used to deduce with (3.4) the boundary condition (3.6)

- Choose the integration volume as show Fig.3.2

$$\int_{v} \nabla\cdot\mathbf{D}\,d\tau = \oint_{surface} \mathbf{D}\cdot d\mathbf{f} = \int_{v} \rho\,d\tau$$

- Since we assumed no free charges: $\rho = 0$ $(\mathbf{D}_{\mathrm{I}} - \mathbf{D}_{\mathrm{II}})\cdot d\mathbf{f} = (D_{n,\mathrm{I}} - D_{n,\mathrm{II}})df = \rho_s df = 0$

We obtain $D_n^{\mathrm{I}} = D_n^{\mathrm{II}}$ (3.6a)

- The same way :

$$\int_{surface} (\nabla\times\mathbf{A})\cdot d\mathbf{f} = \oint_{line} \mathbf{A}\cdot d\mathbf{s}$$

$B_n^{\mathrm{I}} = B_n^{\mathrm{II}}$ (3.6b)

$E_t^{\mathrm{I}} = E_t^{\mathrm{II}}$ (3.7a)

$H_t^{\mathrm{I}} = H_t^{\mathrm{II}}$ (3.7b)

- (3.6) and (3.7) represent the boundary conditions for electric and magnetic fields
- Must be applied to a specific problem:

Using the Maxwell's equation with these set of equations allows one to calculate the properties of the reflected and transmitted beams.

3.1.2 Laws of Reflection and Refraction

- All three beams have the same frequency: $\omega_i = \omega_r = \omega_{tr}$ (3.9)
- The laws of reflection and Snell's law of refraction:

$$\alpha_i = \alpha_r \quad (3.10a)$$

$\mathbf{k}_i, \mathbf{k}_r, e_n$ are in one plane. (3.10b)

- Also reads :

$$\frac{\sin\alpha_i}{\sin\alpha_{tr}} = \frac{n_{\mathrm{II}}}{n_{\mathrm{I}}} \qquad \mathbf{k}_i, \mathbf{k}_r, e_n \text{ are in one plane.} \quad (3.10c)$$

- When $n_{\mathrm{I}} > n_{\mathrm{II}}$, there is a critical angle α_i^c for which $\alpha_{tr} = 90°$

$$\alpha_i^c = \arcsin\frac{n_{\mathrm{II}}}{n_{\mathrm{I}}} \quad (3.11)$$

For $\alpha_i \geq \alpha_i^c$, a totally reflected beam but no transmitted one: total reflection

Fig.3.3a : The distance of a few wavelengths
Fig.3.3b: When thickness of medium Ⅱ ~nm and covered by material Ⅰ: Attenuated ,or frustrated ,total reflection or the optical tunnel effect

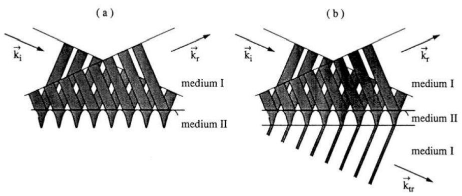


Fig. 3.3. The evanescent wave in the case of total internal reflection (**a**) and the arrangement for the "optical tunneling" effect or attenuated total reflection (**b**)

3.1.4 Reflection and Transmission at an Interface and Fresnel's Formulae

- Weak absorption: $|\kappa| \ll |n|$
- The Fresnel formulae are, for the regime of weak absorption.

$$r_\perp = \frac{n_{\mathrm{I}}\cos\alpha_i - n_{\mathrm{II}}\cos\alpha_{tr}}{n_{\mathrm{I}}\cos\alpha_i + n_{\mathrm{II}}\cos\alpha_{tr}} = -\frac{\sin(\alpha_i - \alpha_{tr})}{\sin(\alpha_i + \alpha_{tr})} \quad (3.18a)$$

$$r_{//} = \frac{-n_{\mathrm{II}}\cos\alpha_i + n_{\mathrm{I}}\cos\alpha_{tr}}{n_{\mathrm{I}}\cos\alpha_{tr} + n_{\mathrm{II}}\cos\alpha_i} = -\frac{\tan(\alpha_i - \alpha_{tr})}{\tan(\alpha_i + \alpha_{tr})} \quad (3.18b)$$

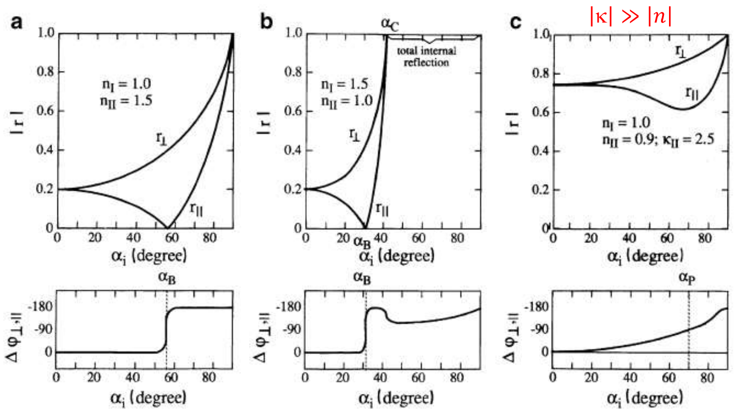
$$t_\perp = \frac{2n_{\mathrm{I}}\cos\alpha_i}{n_{\mathrm{I}}\cos\alpha_i + n_{\mathrm{II}}\cos\alpha_{tr}} = \frac{2\sin\alpha_{tr}\cos\alpha_i}{\sin(\alpha_i + \alpha_{tr})} \quad (3.18c)$$

$$t_{//} = \frac{2n_{\mathrm{I}}\cos\alpha_i}{n_{\mathrm{I}}\cos\alpha_{tr} + n_{\mathrm{II}}\cos\alpha_i} = \frac{2\sin\alpha_{tr}\cos\alpha_i}{\sin(\alpha_i + \alpha_{tr})\cos(\alpha_i - \alpha_{tr})} \quad (3.18d)$$

- From (3.18):

$$r_{//,\perp}(\alpha_i = 0) = \frac{n_{\mathrm{I}} - n_{\mathrm{II}}}{n_{\mathrm{I}} + n_{\mathrm{II}}};\ t_{//,\perp}(\alpha_i = 0) = \frac{2n_{\mathrm{I}}}{n_{\mathrm{I}} + n_{\mathrm{II}}} \quad (3.18e)$$

- Fig.3.6 formulae graphically and the phase shift between the various reflected components.
- For the experimentally accessible quantities *R* and *T*



Strong absorption is not covered by Eq (3.18)

Fig. 3.6 The modulus of the reflection coefficient r for the (electric) field amplitudes according to (3.18) and the phase difference between the reflected amplitudes for the orientations $r_\perp$ and $r_\parallel$, and reflection at an optically thicker (**a**) and thinner (**b**) medium and for reflection at a strongly absorbing medium (**c**) (According to [76P1])

- For the $R_\perp$ increases monotonically with α_i
- For weak absorption $(n >> k)$:

$$R(\alpha_i = 0) = \left(\frac{n_{\mathrm{II}} - n_{\mathrm{I}}}{n_{\mathrm{II}} + n_{\mathrm{I}}}\right)^2 \quad (3.19a)$$

- For $n_I = 1$,again for weak absorption:

$$R(\alpha_i = 0) = \left(\frac{n_{\mathrm{II}} - 1}{n_{\mathrm{II}} + 1}\right)^2 \quad (3.19b)$$

- For strong absorption and $n_I = 1$:

$$R(\alpha_i = 0) = \frac{(n_{\mathrm{II}} - 1)^2 + k_{\mathrm{II}}^2}{(n_{\mathrm{II}} + 1)^2 + k_{\mathrm{II}}^2} \quad (3.19c)$$

- While for grazing incidence in all cases:

$$R(\alpha_i = 90^0) = 1 \quad (3.19d)$$

- In contrast $R_{//}$ goes through a minimum at a certain angle (weak absorption):

$$R_{//}(\alpha = \alpha_B) = 0 \quad (3.20)$$

1.Brewster's angle
2.Only the perpendicularly to the plane of incidence is reflected.
3.Can be used to polarize light.
4.The transmitted beam is not strictly polarized.

- From (3.18b) , when $r_{//} = 0$:

$$n_{\mathrm{II}}\cos\alpha_i = n_{\mathrm{I}}\cos\alpha_{tr} \quad \text{or} \quad \tan(\alpha_i + \alpha_{tr}) = \infty \quad (3.21)$$

- The nontrivial solution: $\alpha_i + \alpha_{tr} = 90°$ (3.22)
- For $n_{\mathrm{II}} < n_{\mathrm{I}}$,the critical angle $\cos\alpha_c$ for total internal reflection exists.

- For strong absorption, R does not reach zero for any polarization .
- The definition of α_p : 1. when the slopes of the curves $R_{//}(\alpha_i)$ and $R_{\perp}(\alpha_i)$ are equal
  2.The minimum of $R_{\perp}$
- For a lossless interface, the power (energy per unit of time): $P_i = P_r + P_{tr}$ (3.23a)
- $|t_{//,\perp}|^2, |r_{//,\perp}|^2$ give the energy flux density.
- The transmitted one as show in Fig.3.7 :

$$T_{//,\perp} = \frac{P_{tr}}{P_i} = \frac{I_{tr}}{I_i}\frac{\cos\alpha_{tr}}{\cos\alpha_i} = |t_{//,\perp}|^2 \frac{\cos\alpha_{tr}}{\cos\alpha_i}$$

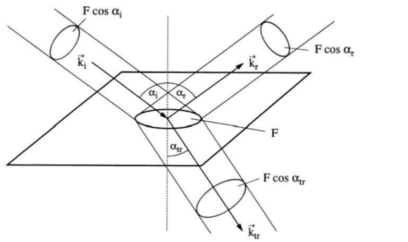


Fig. 3.7. The relation of the cross sections of incident, transmitted and reflected beams at a plane interface between two media

3.1.5 Extinction and Absorption of Light

- Now we discuss on the propagation of a beam within a medium.
- Assumptions: a strictly parallel beam and neglect all diffraction losses .
- The decay usual form: $I = I_0 e^{-\alpha(\omega)d}$ Lambert' law (3.24)
- When the materials active : $\alpha < 0$,while passive : $\alpha > 0$
- The attenuation in 3.24 comprises : $\alpha_{extinction}(\omega) = \alpha_{absorption}(\omega) + \alpha_{scattering}(\omega)$ (3.25)

1.Absorption: transform into like heat , chemical energy or electromagnetic radiation
2.Scattering: The un-shifted component and the frequency shifted parts .

- If the particles are small compared to λ : $\frac{I_{scatter}}{I_{incident}} \propto \omega^4$ (3.26)
  Eg. The sunlight propagating through clear atmosphere .
- If the scattering or absorbing particles are diluted (Beer's law or Lambert-Beer's law)：
  $\alpha(\omega) = n_p \alpha_s(\omega)$ (3.27)
  α_s is the specific extinction constant.

The relationship of the **concentration of a substance** in a media and the **amount of light it absorbs**

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- **3.1.6 Transmission Through a Slab of Matter and Fabry Perot Modes**
- The transmission and reflection of a plane-parallel slab of media of geometrical thickness *d* with ideal, lossless surfaces.
- For strong absorption, easily detect the beam reflected from the front surface:
  $\hat R(\omega) = R(\omega)$ for $\alpha(\omega)d >> 1$

And the most regime : $1 \le \alpha(\omega)d \le 5$

$$\hat T(\omega) = \left[1 - R^{I\to II}(\omega,\alpha_i)\right]\exp\left[-\alpha(\omega)d\cos^{-1}\alpha_{tr}\right] \cdot \left[1 - R^{II\to I}(\omega,\alpha_{tr})\right] \quad (3.20a)$$

$\hat T \approx [1 - R(\omega)]^2 e^{-\alpha(\omega)d}$ for normal incidence (3.20b)

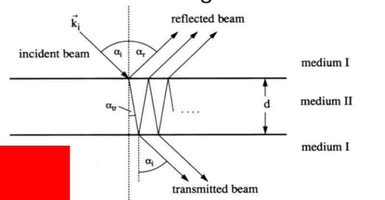


Fig. 3.8. Multiple reflection of an incident light beam in a plane-parallel slab of matter

- Reflection for normal incidence: $\hat R(\omega) \approx R(\omega) + [1 - R(\omega)]^2 R(\omega)\, e^{-2\alpha(\omega)d} \approx R(\omega)$ (3.29c)
- For weak absorption for $\alpha(\omega)d \le 1$, we must consider multiple reflection. 1. For $dl_c^{-1} >> 1$

$$\hat T(\omega) \approx \frac{[1-R(\omega)]^2 e^{-\alpha(\omega)d}}{1 - R(\omega)^2 e^{-2\alpha(\omega)d}} \approx [1-R(\omega)]^2 e^{-\alpha(\omega)d} \quad for \quad \alpha d \le 1 \quad (3.29d)$$

2. For long coherence lengths, form a Fabry-Perot resonator, Case I:

$$\lambda_m = \frac{d\,2n(\omega)}{m} \text{ or } \omega_m = \frac{m\pi c}{dn(\omega_m)} \qquad \delta = n(\omega)k^{vac}d = n(\omega)\frac{\omega d}{c} = m\pi,\ m=1,2,3\ldots \quad (3.30)$$

All partial waves reflected at the two interfaces interfere constructively in the slab,

$$\hat T \to 1, \hat R \to 0$$

Case II: All partial waves interfere destructively in the resonator

$$\hat R \le 1, \hat T << 1$$

- Application: Etalon or Fabry-Perot resonator .
- The general formula for the FP resonator : $\hat T = \frac{A}{1 + F\sin^2\delta}, \hat R = \frac{B + F\sin^2\delta}{1 + F\sin^2\delta}$ (3.31a)

with $F = \frac{4R_\alpha}{(1-R_\alpha)^2}$ $A = \frac{e^{-\alpha d}(1-R_F)(1-R_B)}{(1-R_\alpha)^2}$ $B = \frac{R_F\left(1-\frac{R_\alpha}{R_F}\right)^2}{(1-R_\alpha)^2}$

$R_\alpha = (R_F R_B)^{\frac{1}{2}} e^{-\alpha d}$ $\delta = n(\omega)k^{vac}d = n(\omega)\frac{\omega d}{c}$

Often $R_F = R_B = R$,and when $\alpha d = 0$, we get $A = 1,\ B = 0$

- For $\alpha = 0$ and normal incidence

$$\hat T = \frac{A}{1 + F'\sin^2\delta}, \hat R = \frac{F'\sin^2\delta}{1 + F'\sin^2\delta} = 1 - \hat T \quad (3.32a,b)$$

$$F' = \frac{4R}{(1-R)^2} \quad (3.32c)$$

1. For vanishing damping αd=0, $\hat T(\omega)$ reaches unity and $\hat R(\omega)$ reaches zero
2. Increase F' makes the FP resonances narrower.
3. $\hat T + \hat R = 1$
4. For finite damping, the height of the resonance decreases with increasing αd

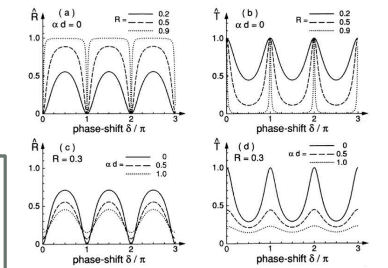


Fig. 3.9. Transmission $\hat T$ (b, d) through and reflection $\hat R$ (a, c) from a Fabry–Perot resonator as a function of the phase shift δ for one half round trip for various values of the reflectivity R of a single surface or the optical density αd of the medium in the resonator

- 3.1.7 Birefringence and Dichroism 不要求
- An anisotropic material: hexagonal wurtzite structure of several semiconductors. In such situations ,the dielectric function $\varepsilon(\omega)$ is generally a tensor.

$$D_i = \varepsilon_0 \sum_j \varepsilon_{ij} E_j; \quad i,j = x,y,z \qquad (3.33)$$

In principle $\varepsilon - tensor$ has nine components. Consider $\varepsilon_{ij} = \varepsilon_{ji}$, left six independent. By a suitable rotation of the cartesian coordinate system, left three elements.

$$\varepsilon_{xx}(\omega) = \varepsilon_{yy}(\omega) \neq \varepsilon_{zz}(\omega) \qquad (3.34a)$$

For biaxial systems of lower symmetry : $\varepsilon_{xx}(\omega) \neq \varepsilon_{yy}(\omega) \neq \varepsilon_{zz}(\omega) \neq \varepsilon_{xx}(\omega)$ (3.34b)

- For cubic crystals: for $\mathbf{k} = 0$: $\varepsilon_{xx}(\omega) = \varepsilon_{yy}(\omega) = \varepsilon_{zz}(\omega)$ (treated as a scalar quantity) for $\mathbf{k} \neq 0$, birefringence and dichroism may appear.
- The consequences tensor character of $\varepsilon(\omega)$ are birefringence and dichroism. See F.3.10

The dichroitic region sometimes covers a wide spectral range, some cases the whole visible part. eg. polaroid films contain long organic molecules.

1. Dichroitic material must show birefringence
2. Birefringent material must have some spectral range in which dichroism occurs.

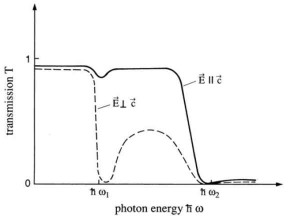


Fig. 3.10. Schematic drawing of the transmission spectra of a dichroitic material for two different polarizations of light with respect to the crystallographic axis

Dichroism means literally that a crystal has two different colors depending on the direction of observation. In a more general sense one describes with the word dichroism every dependence of the absorption spectra on the direction of polarization.

- See Fig.3.11, the ordinary (o) beam and the extraordinary (eo) beam.
- Ordinary (o) beam: perpendicular to the main section.
- extraordinary (eo) beam: has $\boldsymbol{E} \parallel c$ and $\boldsymbol{E} \perp c$
- Application: use birefringent materials as polarizers.
- Negative-uniaxial birefringence: $\Delta n := n_{eo} - n_o < 0$
- Positive-uniaxial birefringence: $\Delta n := n_{eo} - n_o > 0$
- Two simple situations: 1. $\boldsymbol{k} \parallel \boldsymbol{c}$ observes the ordinary beam only
  2. $\boldsymbol{k} \perp \boldsymbol{c}$ By a polarizer, choose the ordinary or extraordinary.

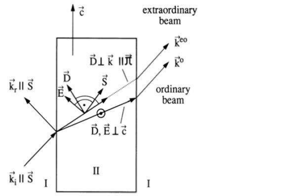


Fig. 3.11. Birefringence for an unpolarized beam falling on a birefringent material at a given angle with the crystallographic axis c parallel to the interface

- Oblique incidence on a surface (see Fig.3.12) are much more complicated.
- Two rather simple ways to present birefringence:

1. The indicatrix
2. The phase velocity in polar coordinates.

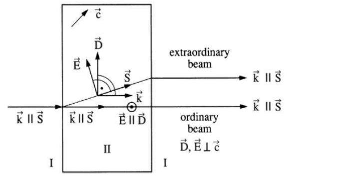


Fig. 3.12. Same as Fig. 3.11 but for normal incidence on a surface cut under an arbitrary angle with respect to c

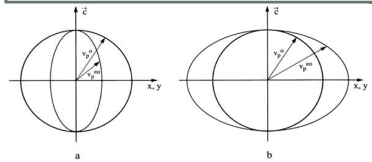


Fig. 3.13. The phase velocity of light in an optically uniaxial material shown for one frequency ω in polar coordinates for positive (a) and negative (b) birefringence for the ordinary (o) and extraordinary (eo) beams

- 3.1.8 Optical Activity （旋光性）
- An optically active material splits a linearly polarized incident beam into a left σ^{-1} and right σ^{+1}circularly polarized one with equal amplitude. Again both components travel with different phase velocities. A superposition after a certain distance results again in linearly polarized light, but with a tilted direction of polarization
- Occurs in amorphous solids including liquids and gases. Also occurs in some crystalline solids: quartz or wurtzite-type crystals .

$$\delta_{\text{rot}} = \varrho(\omega) l \qquad (3.35a)$$

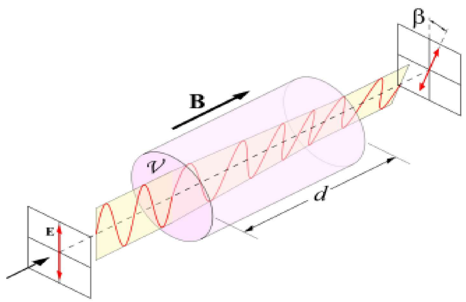
$$\varrho(\omega) = \frac{\pi}{\lambda_0}(n_- - n_+) \qquad (3.35b)$$

$$\begin{pmatrix} D_1 \\ D_2 \\ D_3 \end{pmatrix} = \varepsilon_0 \begin{pmatrix} n^2 & -iG & 0 \\ iG & n^2 & 0 \\ 0 & 0 & n^2 \end{pmatrix} \begin{pmatrix} E_1 \\ E_2 \\ E_3 \end{pmatrix} \qquad (3.35c)$$

$$n_\pm = \sqrt{n^2 \pm G} \qquad (3.35d)$$

- The Faraday effect: The optical activity induced by a magnetic field

$$S_+ = V(\omega) l B \qquad (3.35e)$$



The Faraday effect

- 3.2 Microscopic Aspects 要懂
- **3.2.1 Absorption, stimulated and spontaneous emission, virtual excitation**

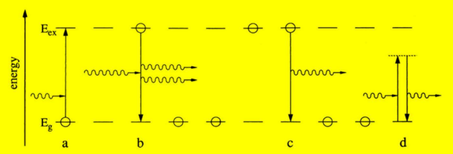


Fig. 3.16. Schematic representations of various interaction processes of light with matter namely absorption (**a**), stimulated emission (**b**), spontaneous emission (**c**) and virtual excitation (**d**)

- Two-level "atoms"

Fig.3.16a: Absorption process $\hbar\omega = E_{\text{ex}} - E_g$

Fig.3.16b: Stimulated emission,
Basic mechanism for all lasers (light amplification by stimulated emission of radiation).

Fig.3.16c: Spontaneous emission, an electron in the excited state transit to the ground state by itself.

Fig.3.16d: Virtual excitation: with an energy different from the eigen-energy of this excited state.
Uncertainty principle of quantum mechanics :

$$\Delta x_i \Delta p_i \geq h \quad for \quad i = 1,2,3. \qquad (3.37a)$$

$$\Delta E \Delta t \approx \hbar \qquad (3.37b)$$

From 3.37b , it is possible to violate energy conservation by an amount ΔE up until a maximum time Δt .

- 3.~~2~~.2 Perturbative Treatment of the Linear Interaction of light with Matter 不用

The Hamiltonian of the system elaborating the perturbation terms.

How a perturbation causes transitions between various eigenstates.

Join these two things together ending with an understanding of the theoretical description of absorption and stimulated or spontaneous transitions.

- The Hamiltonian of the total system: $H = H_{el} + H_{rad} + H_{interac}$ (3.40)
- Using semiclassical treatment of radiation. We treat the radiation field as a small perturbation.
- Replacing the conjugate momentum $\mathbf{p}$ by: $\mathbf{p} \to \mathbf{p} - e\mathbf{A}$

  if we replace $\mathbf{p}$ by its operator: $\mathbf{p} = \frac{\hbar}{i}\nabla$

  then the single particle Hamiltonian reads: $H = \frac{1}{2m}\left(\frac{\hbar}{i}\nabla - e\mathbf{A}\right)^2 + V(\mathbf{r})$ (3.43)

  Using the Coulomb gauge : $H = \frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}) - \frac{e}{m}\mathbf{A}\frac{\hbar}{i}\nabla + \frac{e^2}{2m}\mathbf{A}^2$

$$H = H_{el} - \frac{e}{m}\mathbf{A}\frac{\hbar}{i}\nabla + \frac{e^2}{2m}\mathbf{A}^2 \qquad H = H_{el} + H^{(1)} + H^{(2)}$$

- From the above, if we assume the light intensity are small, $H^{(1)}$is a perturbation term of first order and $H^{(2)}$is small of second order.
- Fermi's golden rule for the transition rate ω_{ij} :

  Time-dependent Schrodinger equation : $H\psi = i\hbar\frac{\partial\psi}{\partial t}$

  with the stationary solutions when $H = H_0$ $\psi_n(\mathbf{r},t) = \varphi_n(\mathbf{r})e^{-i\left(\frac{E_n}{\hbar}\right)t}$ (3.47b)

  For with a perturbation $H^{(1)}$:

$$\psi(\mathbf{r},t) = \sum_n a_n(t)\varphi_n(\mathbf{r})e^{-i\left(\frac{E_n}{\hbar}\right)t} \qquad (3.48)$$

  Assume the perturbation is switched on at t=0. And:

$$\left.\begin{array}{l} a_i(t) = 1 \\ a_{n\neq i}(t) = 0 \end{array}\right\} \quad for \quad t \leq 0 \qquad (3.49)$$

  For t>0 , transition rate-Fermi golden rule

$$\omega_{ij} = \frac{2\pi}{\hbar}\left|H_{ij}^{(1)}\right|^2 D(E) \qquad (3.50a)$$

  The transition matrix element given by:

$$H_{ij}^{(1)} = \int \psi_j^*(\mathbf{r})H^{(1)}\psi_i(\mathbf{r})d\tau = \langle\psi_j|H^{(1)}|\psi_i\rangle \qquad \omega_{ij} \propto \left|H_{ij}^{(1)}\right|^2 \qquad (3.50b)$$

- If $H_{ij}^{(1)} = 0$, then the second-order contribution reads：

$$\omega_{ij}^{(2)} = \frac{2\pi}{\hbar}\left|\sum_{k\neq i,j}\frac{H_{jk}^{(1)}H_{ki}^{(1)}}{E_i - E_k} + H_{ij}^{(2)}\right|^2 D(E) \qquad (3.51)$$

- Some more detail for the perturbations:

  with the vector potential $\mathbf{A}$: $\mathbf{A} = \mathbf{A}_0 e^{i(\mathbf{kr}-\omega t)}$ (3.52a)

  we find , e.g., for the absorption process (3.50):

$$\omega_{g\to ex} = \frac{2\pi}{\hbar}\left|\frac{-e\hbar}{im}A_0 \cdot \int \varphi_{ex}^*(\boldsymbol{r})e^{i(E_{ex}/\hbar)t}\boldsymbol{e}_A e^{i(\boldsymbol{kr}-\omega t)}\nabla\varphi_g(\boldsymbol{r})e^{-i(E_g/\hbar)t}d\tau\right|^2 D(E_g + \hbar\omega)$$

$$\propto \left|A_0\langle\varphi_{ex}|H^{(1)}|\varphi_g\rangle\right|^2 D(E_g + \hbar\omega) =:$$

$$A_0^2\left|H_{eg}^{(1)}\right|^2 D(E_g + \hbar\omega) \qquad (3.52b)$$

  A significant transition rate occurs only if the time dependent vanish :

$$E_{ex} - E_g - \hbar\omega = 0 \qquad (3.53a)$$

  If the φ_i have a wave vector $\mathbf{k}$: $\hbar\mathbf{k}_{ex} - \hbar\mathbf{k}_g - \hbar\mathbf{k} = 0$ (3.53b)

  so : $\omega_{ij} \propto A_0^2\left|H_{eg}^{(1)}\right|^2 \propto I\left|H_{eg}^{(1)}\right|^2 \propto N_{ph}\left|H_{eg}^{(1)}\right|^2$ (3.54)

  Because of the orthonormal set: $\left|\int \varphi_j^* H^{(1)}\varphi_i d\tau\right|^2 = \left|\int \varphi_i^* H^{(1)}\varphi_j d\tau\right|^2$ (3.55)

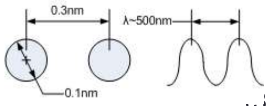
  So we find that:

The probabilities for induced emission and absorption are the same and that the rates differ only by factors containing the number of atoms in the upper and lower states.

- Simplify the interaction operator $H^{(1)}$:

The dipole approximation:

Thus expand $e^{i\mathbf{kr}}$ in (3.52) : $e^{i\mathbf{kr}} = 1 + \frac{i\mathbf{kr}}{1!} + \frac{i\mathbf{kr}^2}{2!} + \cdots \simeq 1.$ (3.56)



Also：

$$\hbar\mathbf{k}_{ex} - \hbar\mathbf{k}_g - \hbar\mathbf{k} = 0 \rightarrow \hbar\mathbf{k}_{ex} - \hbar\mathbf{k}_g = 0$$

with : $H_{ij}^{(1)} \sim \langle\varphi_i|\mathbf{p}|\varphi_i\rangle = \langle\mathbf{p}_{i,j}\rangle.$ $\quad \frac{\hbar}{i}\nabla = p = m\dot{\mathbf{r}}$

so we find that : $\int \varphi_j^* \frac{\hbar}{i}\nabla\varphi_i d\tau = m\frac{i}{\hbar}(E_i - E_j)\int \varphi_j^* \mathbf{r}\varphi_i d\tau = m\omega \int \varphi_j^* \mathbf{r}\varphi_i d\tau$ (3.58)

Then the transition rate using this so-called dipole approximation :

$$\omega_{ij} \sim I\omega^2 |e_A\langle e\mathbf{r}_{ij}\rangle|^2 D(E) = I\omega^2 |H_{ij}^D|^2 D(E) \qquad (3.59)$$

$|H_{ij}^D|^2$: the dipole-transition probability .

- To conclude this section: the net rate in Fig.3.16a-c : 可能考计算，速率方程

$$\frac{\partial N_{ph}}{\partial t} = \boxed{-N_A\alpha_g N_{ph}|H_{g\rightarrow e}^D|^2} \boxed{+ N_A(1-\alpha_g)(1+N_{ph})|H_{e\rightarrow g}^D|^2} \qquad (3.61)$$

Absorption Spontaneous and stimulated emission

In which α_g is a fraction in the ground state, $(1-\alpha_g)$ are in the excited state.

From (3.55) : $|H_{g\rightarrow e}^D|^2 = |H_{e\rightarrow g}^D|^2 = |H^D|^2$ (3.62)

and hence:

$$\frac{1}{|H^D|^2}\frac{\partial N_{ph}}{\partial t} = \boxed{N_{ph}N_A(1-2\alpha_g)} + \boxed{N_A(1-\alpha_g)} \qquad (3.63)$$

→ Linearly depends on N_{ph}
Absorption and stimulated emission

→ Independent of N_{ph}
Spontaneous emission

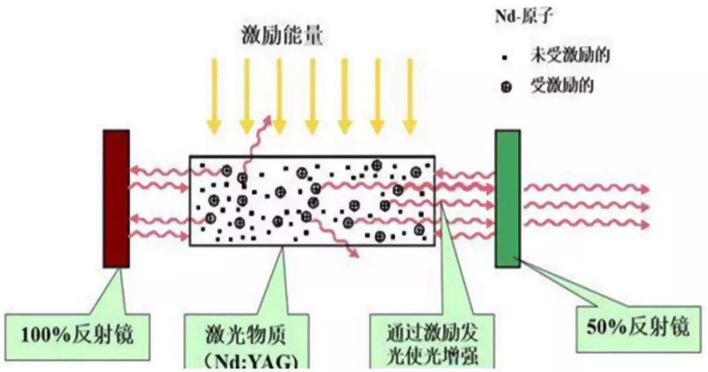
1. For net absorption : $\alpha_g > \frac{1}{2}$
2. Amplification or optical gain : $\alpha_g < \frac{1}{2}$
3. The semiconductor becomes transparent $\alpha_g = \frac{1}{2}$
4. The situation $\alpha_g < \frac{1}{2}$ can't be reached in thermal equilibrium, but only under a pump source

- Elucidated : $\alpha_g(t=0) = 1$, we have absorption : $-\alpha(\omega) \sim N_A(1-2\alpha_g)$ (3.64)

so : $t\uparrow, pump \quad power\uparrow \Rightarrow \alpha_g\downarrow \Rightarrow \alpha(\omega)\downarrow$

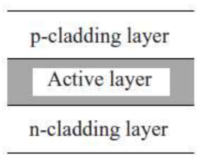
when $\alpha_g = \frac{1}{2}$, the absorption vanishes and the material becomes transparent.

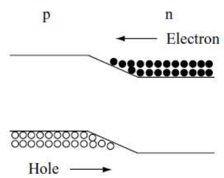
Further increase the pump, light amplification achieved.



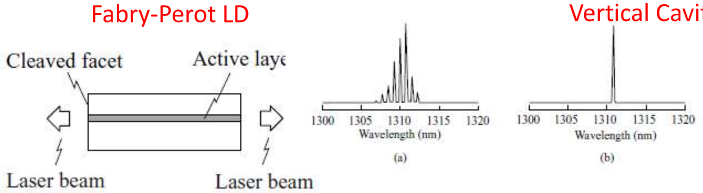
各种类型半导体激光器

了解

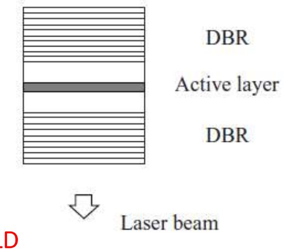




Fabry-Perot LD



Vertical Cavity Surface Emitting Laser



DFB LD

DBR LD